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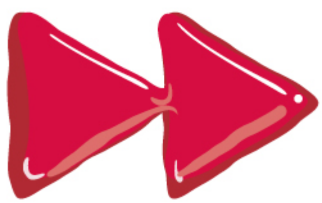

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PBM, patient blood management. **References: 1.** Leahy MF *et al. Transfusion.* 2017;57:1347–58. **2.** Althoff FC *et al. Ann Surg.* 2019;269:794–804

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Original Article

A randomised, controlled trial evaluating a low cost, 3D-printed bronchoscopy simulator*

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Summary

Low-fidelity, simulation-based psychomotor skills training is a valuable first step in the educational approach to mastering complex procedural skills. We developed a cost-effective bronchial tree simulator based on a human thorax computed tomography scan using rapid-prototyping (3D-print) technology. This randomised, single-blind study evaluated how realistic our 3D-printed simulator would mimic human anatomy compared with commercially available bronchial tree simulators (Laerdal® Airway Management Trainer with Bronchial Tree and AirSim Advance Bronchi, Stavanger, Norway). Thirty experienced anaesthetists and respiratory physicians used a fibreoptic bronchoscope to rate each simulator on a visual analogue scale (VAS) (0 mm = completely unrealistic anatomy, 100 mm = indistinguishable from real patient) for: localisation of the right upper lobe bronchial lumen; placement of a bronchial blocker in the left main bronchus; aspiration of fluid from the right lower lobe; and overall realism. The 3D-printed simulator was rated most realistic for the localisation of the right upper lobe bronchial lumen ($p = 0.002$), but no differences were found in placement of a bronchial blocker or for aspiration of fluid ($p = 0.792$ and $p = 0.057$) compared with using the commercially available simulators. Overall, the 3D-printed simulator was rated most realistic ($p = 0.021$). Given the substantially lower costs for the 3D-printed simulator (£85 (€100/US$110) compared with > ~ £2000 (€2350/US$2590) for the commercially available simulators), our 3D-printed simulator provides an inexpensive alternative for learning bronchoscopy skills, and offers the possibility of practising procedures on patient-specific models before attempting them in clinical practice.

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Introduction

Low-fidelity, simulation-based psychomotor skills training is a valuable educational approach for learning complex procedural skills [1, 2]. Several authors have reported that low-fidelity airway simulators are as effective in learning basic psychomotor skills as high-fidelity simulators [3–7]. Flexible bronchoscopy, a core competence in anaesthesia [8], intensive care medicine [9] and respiratory medicine [10], is an invasive procedure for which simulation is a well-established training method [11, 12].

High-quality education in flexible bronchoscopy is facilitated by the availability of reasonably priced, anatomically-correct simulators. Commercially available bronchoscopy simulators cost between £1500 and £3000 (€1800 to €3.500/$1800 to $3700) [13, 14]. Rapid-prototyping technology (3D-printing) is becoming more and more available [15] for medical and educational purposes. We printed a low-fidelity bronchial tree simulator, based on human thorax computed tomography (CT) scans, at costs of less than 10% of the commercially available simulators.

The purpose of this study was to evaluate the validity of the 3D-printed simulator by comparing its anatomical realism to commercially available bronchial tree simulators. For that purpose, the study participants had to perform several bronchoscopic tasks; we chose correct anatomical identification of the right upper lobe bronchial lumen as our primary outcome, as this is a crucial first step for performing effective fibrescopic examination of the bronchial tree.

Methods

The Cantonal Ethics Committee of Bern, Switzerland evaluated our single-blind, randomised, controlled study. Eighteen consultant anaesthetists and 12 consultant respiratory physicians from Bern University Hospital, who perform fibreoptic procedures regularly several times a week, were enrolled in 2016, all of who provided written informed consent for participation.

This study compared three different bronchial tree simulators (Fig. 1): a Laerdal® Airway Management Trainer with Bronchial Tree (Laerdal, Stavanger, Norway) (‘Laerdal’) [16, 17]; an AirSim Advance Bronchi (TruCorp® Ltd, Belfast, Northern Ireland) (‘TruCorp’) [18]; and our solid 3D-printed simulator (‘3D-print’) [19], based on CT scans of the thorax from a volunteer [20]. The 3D-printed bronchial tree was attached to the head of a Laerdal Airway Management Trainer to ensure blinding. For blinding purposes, we placed the three simulators in cardboard boxes and hid them under surgical aperture drapes (Fig. 2). The trachea of each simulator was intubated using an ID 8.0-mm tracheal tube (Rüsch®, Teleflex Incorporated, Limerick, PA, USA), with a measured distance of 4 cm between the end of the tube and the tracheal carina. All tubes were cuffed and fixated at the heads with the Ambu® ET Tube Holder (Ambu®, Ballerup, Denmark).

Participants were asked to perform three different procedures on each simulator: bronchoscopic identification of the right upper lobe bronchial lumen; placement of a bronchial blocker (Uniblocker™, Fuji Systems Corporation, Tokyo, Japan) in the left main bronchus; and aspiration of 10 ml of tap water from the right lower lobe. Participants used an Ambu

Figure 1 Simulators used. From the left: Laerdal® Airway Management Trainer with Bronchial Tree, AirSim Advance Bronchi, and the 3D-printed simulator connected to a Laerdal® Head.

Figure 2 Blinded study setup. The three simulators covered with aperture drapes and the fibreoptic bronchoscope.

aScope™ 3 Slim 3.8/1.2 bronchoscope with an Ambu aView™ monitor, and a Laerdal Suction Unit LSU, equipment with which they were familiar through clinical use.

A random allocation sequence was generated using www.randomizer.org to randomly assign the order of the tasks and the simulators for each participant.

Participants rated the realism directly after performing each task. At the end of the study, each participant rated the overall realism of each simulator. The primary outcome was the realism of the bronchoscopic identification of the right upper lobe bronchial lumen rated using a 100 mm visual analogue scale (VAS), where '0 mm' represented 'completely unrealistic anatomy', and '100 mm' represented 'indistinguishable from a human being'. The secondary outcomes were: how realistic the placement of a bronchial blocker in the left main bronchus was; how realistic the aspiration of 10 ml tap water from the right lower lobe was; and how realistic the simulators were in general, compared with a human being. These secondary outcomes were rated on the same 100 mm VAS as the primary outcome.

We also recorded the time needed for each procedure (measured from insertion of the fibreoptic scope into the tracheal tube until completion of the task in seconds), specialisation, years of experience, sex, and age of the participating physicians.

We hypothesised our 3D-printed simulator would be non-inferior for our primary outcome compared with the commercially available simulators. We assumed that the commercially available simulators would be rated with 60 mm on the VAS compared with a human being (100 mm) for our primary outcome. We defined non-inferiority as the 3D-printed simulator remaining within a mean (SD) 10 (15) mm of the VAS compared with the commercially available devices. Sample size was calculated with a one-sided significance level of 0.05, thus requiring 16 subjects to reach a power of 80%. To account for these assumptions and to secure the power of the study, we aimed to include 30 participants.

We compared the three devices with Friedman's test and performed post-hoc pairwise comparisons with Wilcoxon's signed-rank test with Bonferroni correction for multiple comparisons. A probability of less than 0.05 was considered as statistically significant. All statistics were calculated using Stata/SE 14.1 (Stata Corp. LP, College Station, TX, USA).

Results

Thirty participants completed the study, with no dropouts (25 male, 5 female, median (IQR [range]) age 41 (36–48 [30–60]) years and time medically qualified 12 (10–18 [5–32]) years).

Visual analogue scale (VAS) scores (mm) for simulator realism are shown in Table 1. The 3D-printed simulator was rated significantly more anatomically realistic by participants than the commercially available simulators for our primary outcome, and non-inferior in terms of bronchial blocker placement in the left main bronchus and aspiration of fluid from the right lower lobe bronchus. The 3D-printed simulator was rated overall significantly more realistic compared with the two commercially available simulators.

Times taken for each task are shown in Table 2. These results suggest that the right upper lobe bronchus can be identified quicker using the 3D-printed simulator, and a bronchial blocker can be placed quicker using the TruCorp simulator, compared with the other two simulators on trial.

Discussion

Thirty hospital consultants in anaesthesia and respiratory medicine rated our 3D-printed bronchoscopy simulator non-inferior to two commercially available

Table 1 Visual analogue scale (VAS) score (mm) for the realism of all four outcome parameters. Values are median (IQR [range]).

Outcome	Laerdal	TruCorp	3D-print	p value
Localisation of right upper lobe bronchial lumen	65 (43–80) [18–99]	68 (32–76) [17–99]	77 (65–88) [48–99]	0.002†
Placement of bronchial blocker in the left main bronchus	70 (56–82) [18–89]	73 (63–81) [19–99]	69 (51–81) [41–94]	0.792
Aspiration of fluid from the right lower lobe	53 (39–70) [17–86]	62 (48–71) [3–87]	64 (52–81) [13–89]	0.057
Overall realism	63 (51–77) [9–93]	66 (53–78) [8–93]	75 (68–83) [48–99]	0.021‡

†P = 0.003 for 3D-print vs. TruCorp, p = 0.658 for TruCorp vs. Laerdal, and p = 0.007 for 3D-print vs. Laerdal.
‡p = 0.044 for 3D-print vs. TruCorp, p = 0.673 for TruCorp vs. Laerdal, and p = 0.004 for 3D-print vs. Laerdal.

Table 2 Time (s) taken to complete each of the three tasks (non-sequentially) in the three simulators. Values are median (IQR [range]).

Outcomes	Laerdal	TruCorp	3D-print	p value
Localisation of right upper lobe bronchial lumen	16 (12–22) [7–84]	17 (13–23) [8–57]	12 (9–17) [7–71]	0.009†
Placement of bronchial blocker in the left main bronchus	81 (49–164) [34–1061]	43 (32–56) [23–76]	88 (53–151) [27–864]	< 0.001‡
Aspiration of fluid from the right lower lobe	91 (76–124) [31–184]	87 (68–123) [38–153]	92 (64–128) [40–192]	0.648

†p = 0.039 for 3D-print vs. TruCorp, p = 0.967 for TruCorp vs. Laerdal and p = 0.02 for 3D-print vs. Laerdal.
‡p < 0.001 for 3D-print vs. TruCorp, p = 0.688 for 3D-print vs. Laerdal and p < 0.001 for TruCorp vs. Laerdal.

bronchoscopy simulators from Laerdal and TruCorp, in terms of the anatomical realism of three standardised bronchoscopic tasks.

The 3D-printed simulator was rated significantly more anatomically realistic overall and for identifying the right upper lobe bronchial lumen than the commercially available simulators. Such identification is important for training operators to orientate the bronchoscope correctly within the bronchial tree.

The placement of bronchial blockers allows separation of the lungs in the management of haemorrhage or for one-lung ventilation [21]. We demonstrated that the 3D-printed simulator provides similar anatomical realism to commercial simulators from Laerdal and TruCorp, so could be used effectively for teaching this skill. Many participants observed that the haptic aspects of being able to place a blocker during simulation were of great importance, although we did not evaluate this.

Although no significant difference was found between the simulators, aspiration of fluid from the right lower lobe was rated as the least realistic simulation, because subjectively the aspirate was thought not to resemble the fluids usually found in the lungs of patients. However, fluids other than the tap water we used would have damaged the commercially available simulators over time, because they are difficult to clean. De novo manufacture of the 3D-printed simulators or postprint modification could allow for use of other fluids.

Our study is the first to compare a 3D-printed bronchoscopy simulator with commercially available bronchoscopy simulators, although 3D-printed bronchoscopy simulators have been recently assessed for training purposes [22, 23]. As the technology becomes more easily accessible [15], it offers the possibility of extending access to basic bronchoscopy simulation training, by lowering the cost involved and increasing availability. The 3D-printed simulator cost £85 (€100/US$110) to produce, whereas the Laerdal Airway Management Trainer with Bronchial Tree costs £1996 (€2390/US$2570, excluding VAT), and the AirSim Advance Bronchi £2490 (€2990/US$3200, excluding VAT).

At present, only commercially available manikins can simulate tracheal intubation simultaneously, but it should be possible in the near future to scan an airway and 3D print a model of it using a realistic polymer.

We accept that our study has several limitations. We could not perform a double-blind study, since the study personnel had to prepare the simulators and instruct participants in which task to perform on what simulator. However, instructions were given to participants simply to evaluate three different bronchoscopy simulators, without the instructors inferring that any of the three simulators was not commercially available. Secondly, the participants did not rate the anatomical realism of each simulator directly to human anatomy, but to their memory of human anatomy. Participants were experienced in flexible bronchoscopy, and performed this task several times a week, but our results should be interpreted as indicating relative realism between simulators rather than providing a quantitative absolute comparison between simulation and clinical reality. Related to this, the task times quoted appertain only to the simulation, and cannot be compared with published clinical task times. Furthermore, the study was not powered to detect the significance of differences in task times, so these results can only be interpreted as indicative of our 3D print's usefulness as a simulator in comparison with the commercially available simulators assessed.

In conclusion, our 3D-printed bronchoscopy simulator compares favourably with two commercially available bronchoscopy simulators, but for ~ 10% of the costs. 3D-printed simulation is an inexpensive alternative for teaching bronchoscopy skills and offers the possibility of simulating specific patients' anatomy in vitro, and planning subsequent in-vivo bronchoscopy accordingly.

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